

## Project Initialization and Planning Phase

|               |                                                                          |
|---------------|--------------------------------------------------------------------------|
| Date          | 19 June 2025                                                             |
| Team ID       | SWTID1749908722                                                          |
| Project Name  | CardMaster: Intelligent Playing Card Recognition using Transfer Learning |
| Maximum Marks | 3 Marks                                                                  |

## Project: CardMaster – Intelligent Playing Card Recognition using Transfer Learning

In today's tech-driven world, users across gaming, surveillance, and education domains seek smarter ways to bridge physical and digital interactions with playing cards.

**CardMaster** aims to solve this by developing a **high-accuracy model for recognizing playing cards** using **transfer learning**. Leveraging a dataset of **8,154 cropped images** across **53 card classes**, the system ensures **efficient feature extraction**, **reduced computational costs**, and **real-time recognition**.

- In **digital card games**, players struggle to sync physical cards with digital platforms due to the absence of automated recognition—leading to a fragmented experience.
- In **casinos**, surveillance teams find it difficult to detect cheating in real-time, lacking automated systems to track card movement and game integrity.
- In **interactive learning**, students lack tools that respond to physical card play with instant feedback, limiting hands-on learning opportunities.

CardMaster addresses these needs by enabling real-time, intelligent card recognition across platforms—**enhancing user experience, security, and learning outcomes**.

### CardMaster—Intelligent Playing Card Recognition Using Transfer Learning

|                     |                                                                     |
|---------------------|---------------------------------------------------------------------|
| I am                | A digital card game player                                          |
| am trying to        | Play physical card games that automatically sync and update         |
| but                 | My digital system does not recognize the physical cards I'm playing |
| because             | There is no real-time, high-accuracy playing card recognition model |
| which makes me feel | Frustrated and disconnected from modern gaming experiences          |

### Example:

| I am :                                          | I'm Trying to                                               | But                                                        | Because                             | Which makes me feel      |
|-------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|-------------------------------------|--------------------------|
| A student learning to play strategic card games | get real-time guidance based on the cards I play physically | my training tool doesn't recognize real cards on the table | it only supports digital simulation | unsupported and confused |

### Customer Problem Statement Table

| Problem Statement (PS) | I am (Customer)                                   | I'm trying to                                                  | But                                                               | Because                                                                | Which makes me feel                           |
|------------------------|---------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------|
| PS-1                   | A digital card game player                        | synchronize my physical card gameplay with a digital interface | the system does not recognize physical cards in real time         | there's no intelligent camera-based recognition integrated             | disconnected and frustrated                   |
| PS-2                   | A casino surveillance officer                     | monitor each card dealt during high-stakes games               | I have to rely on manual observation and video reviews            | automated card recognition tools are not readily available or reliable | overwhelmed and concerned about game fairness |
| PS-3                   | A student learning to play strategic card games   | get real-time guidance based on the cards I play physically    | my training tool doesn't recognize real cards on the table        | it only supports digital simulations                                   | unsupported and confused                      |
| PS-4                   | A developer building smart card game applications | integrate a card recognition feature easily                    | existing models require training from scratch and are data-hungry | there's no pre-trained, plug-and-play solution available               | restricted and less innovative                |